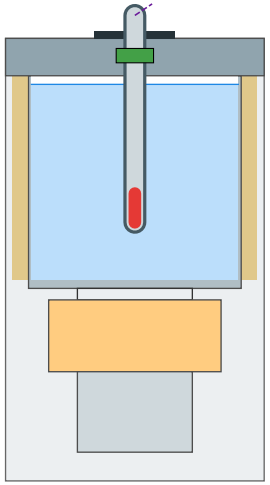
ThermoX - 10 THERMOWELL POSITIONS (mock designs)

Round bottle, cut through the middle. Grey tube = thermowell (8 mm stainless, closed end), red = sensor, green = seal, purple dashed = cable.

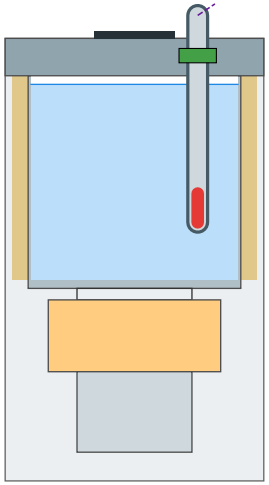
1



Lid centre

score 17/25

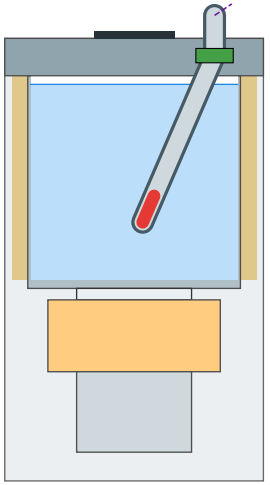
2



Lid off-centre

score 20/25

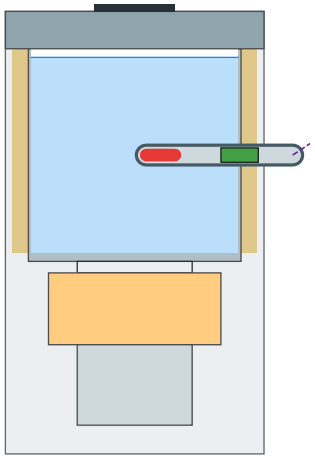
3



Lid angled

score 19/25

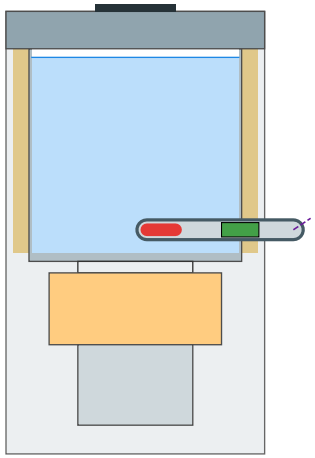
4



Wall middle

score 17/25

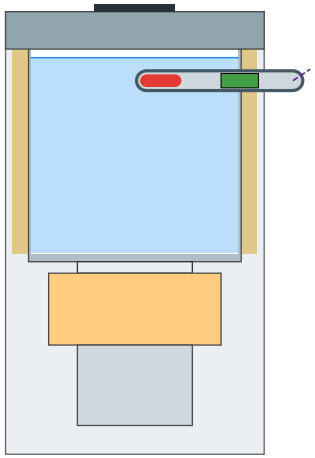
5



Wall low

score 15/25

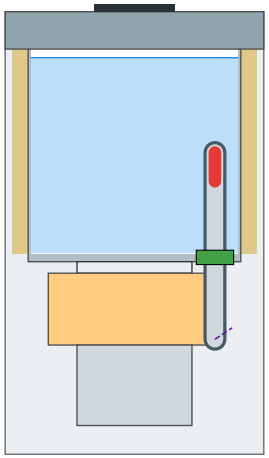
6



Wall high

score 15/25

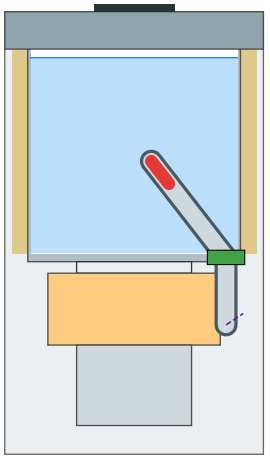
7



Floor up

score 12/25

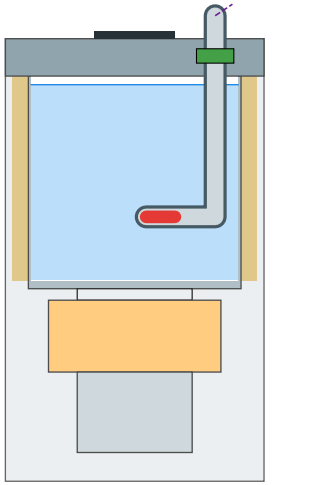
8



Floor angled

score 11/25

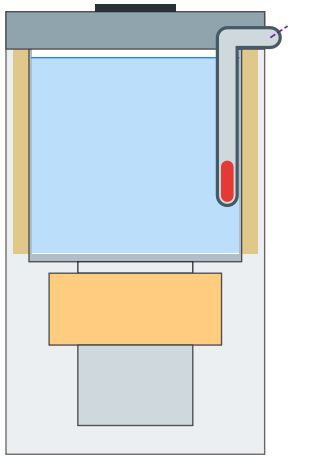
9



Lid L-shape

score 17/25

10



Rim, on the wall

score 22/25

Scores (1-5 each): Accuracy, No leak, Cleaning / drinking, Cable / lid, Easy to build. Recommended: design 10, then design 2. Page 2: comparison. Pages 3-12: each design. Page 13: how to build the thermowell.

COMPARISON (5 = best)

Design	Where	Accuracy	No leak	Cleaning / drinking	Cable / lid	Easy to build	Total
1	Lid centre	*****	*****	**...	***..	**...	17/25
2	Lid off-centre	****.	*****	****.	***..	****.	20/25
3	Lid angled	*****	*****	***..	***..	***..	19/25
4	Wall middle	****.	**...	****.	*****	**...	17/25
5	Wall low	**...	**...	****.	*****	**...	15/25
6	Wall high	*....	***..	****.	*****	**...	15/25
7	Floor up	****.	*....	****.	**...	*....	12/25
8	Floor angled	****.	*....	***..	**...	*....	11/25
9	Lid L-shape	*****	*****	**...	***..	**...	17/25
10	Rim, on the wall	****.	*****	****.	*****	****.	22/25

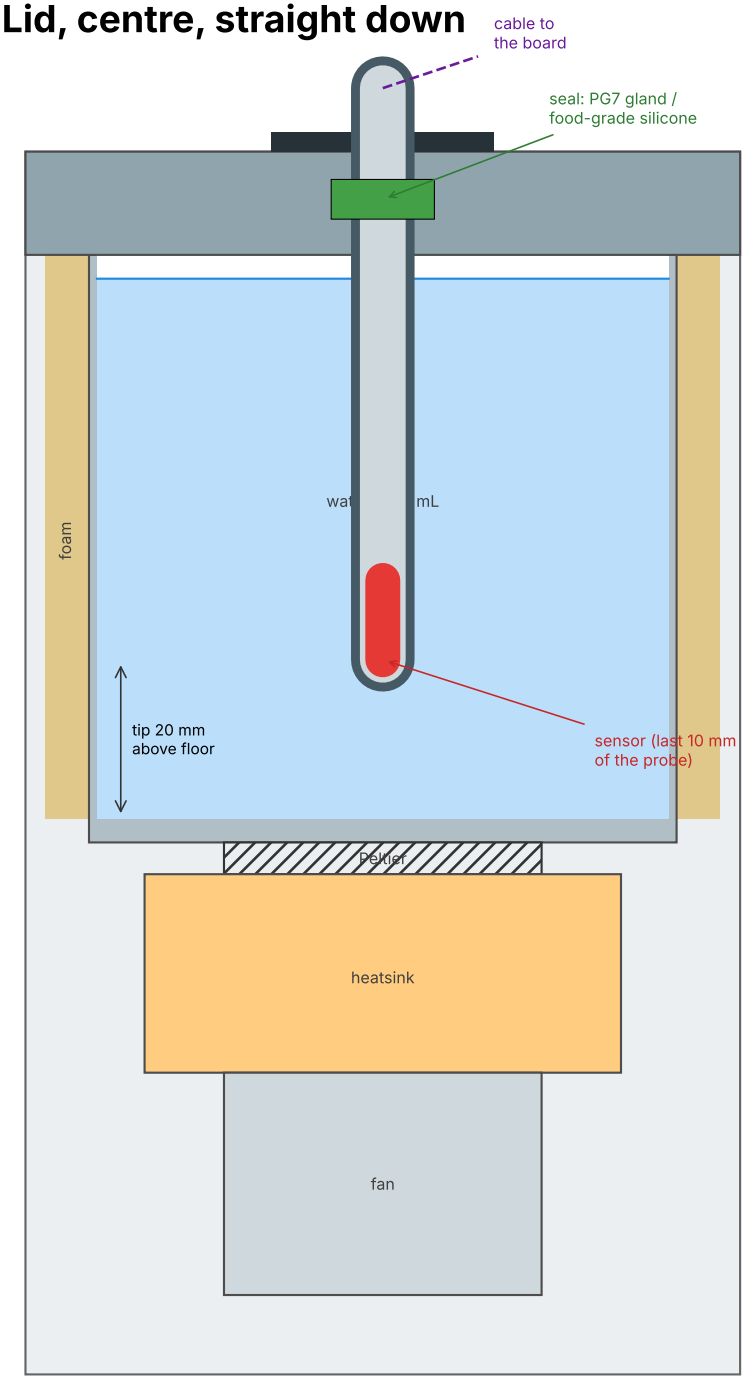
How the scores were given (our judgement, to be checked with tests):

- Accuracy: does the sensor sit in water that stands for the whole cup (not next to the Peltier, not at the surface)?
- No leak: a hole below the water line has to be sealed and must stay food-safe; no hole scores 5.
- Cleaning / drinking: is the tube in the way when you drink, pour or wash the cup?
- Cable / lid: the OLED is already in the lid; a sensor in the lid needs a plug to take the lid off.
- Easy to build: straight tube and one gland score high; holes in the 1-3 mm aluminium cup score low.

Recommendation: design 10 (no hole, lid free, easy). Second choice: design 2 (in the lid, centre free).

Whatever you choose, prove it with the validation test: compare with a reference thermometer at 10-50 C,
measure the response time with and without thermal paste, and repeat each test 3-5 times.
Rule of thumb for thermowells: immersion about 10 x the tube diameter (here ~80 mm). The cup is only 68 mm deep,
so keep the tip low (about 20 mm above the floor), use thermal paste, and measure the lag instead of guessing it.

DESIGN 1. Lid, centre, straight down



What it is

A straight tube hangs from the middle of the lid. The tip sits 20 mm above the floor, in the middle of the water.

Good

+ Reads the bulk water in the centre
+ No hole below the water line
+ The OLED wires already go to the lid

Not so good

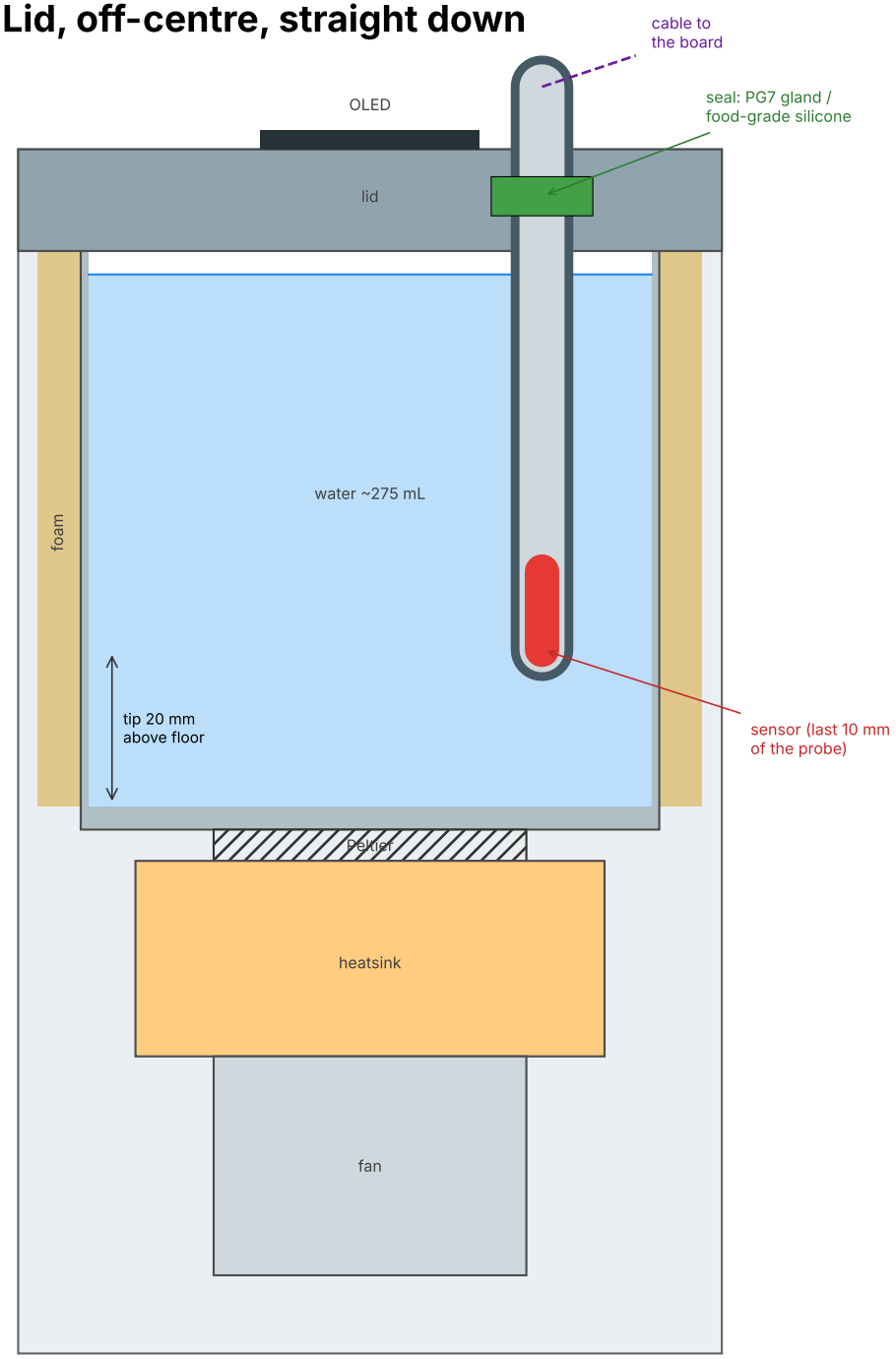
- The OLED sits in the middle of the lid: it would have to move
- The tube is in the way when drinking and cleaning
- The lid needs a plug for the sensor cable

Score

Accuracy	*****
No leak	*****
Cleaning / drinking	**...
Cable / lid	***..
Easy to build	**...
Total	**17/25**

Mock design for comparison: sizes from the round bottle (cup 72 mm inside, water 68 mm deep). Not built or tested yet.

DESIGN 2. Lid, off-centre, straight down



What it is

Same as design 1, but 22 mm from the centre, so the middle of the cup stays free for drinking.

Good

+ No hole below the water line
+ Centre stays free
+ Simple: one gland in the lid

Not so good

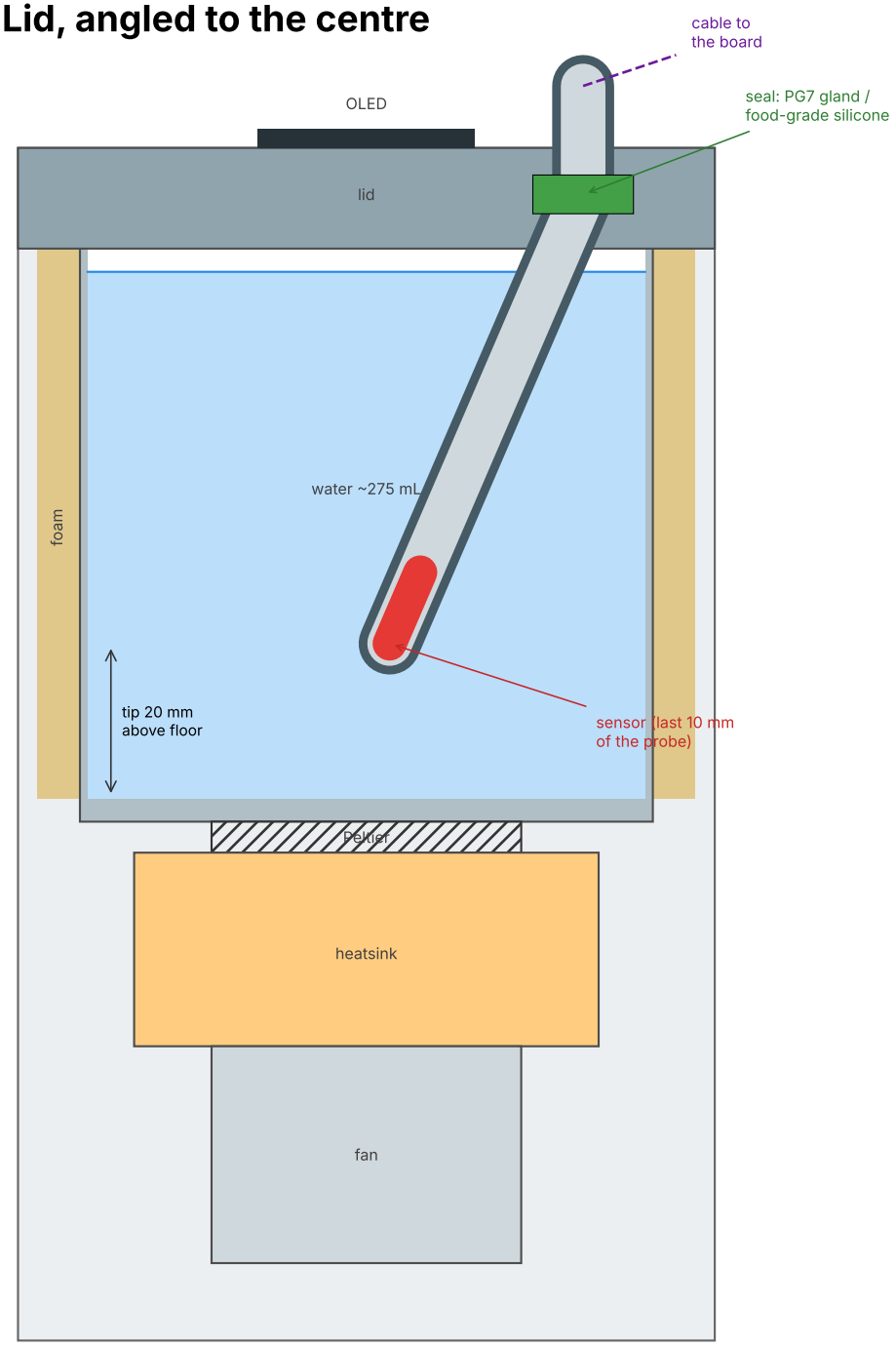
- A little closer to the wall than design 1
- The lid needs a plug for the sensor cable

Score

Accuracy	****.
No leak	*****
Cleaning / drinking	****.
Cable / lid	***..
Easy to build	****.
Total	**20/25**

Mock design for comparison: sizes from the round bottle (cup 72 mm inside, water 68 mm deep). Not built or tested yet.

DESIGN 3. Lid, angled to the centre



What it is

Enters near the lid edge and slopes so the tip ends at the centre of the water.

Good

+ Tip in the centre, entry out of the way
+ No hole below the water line

Not so good

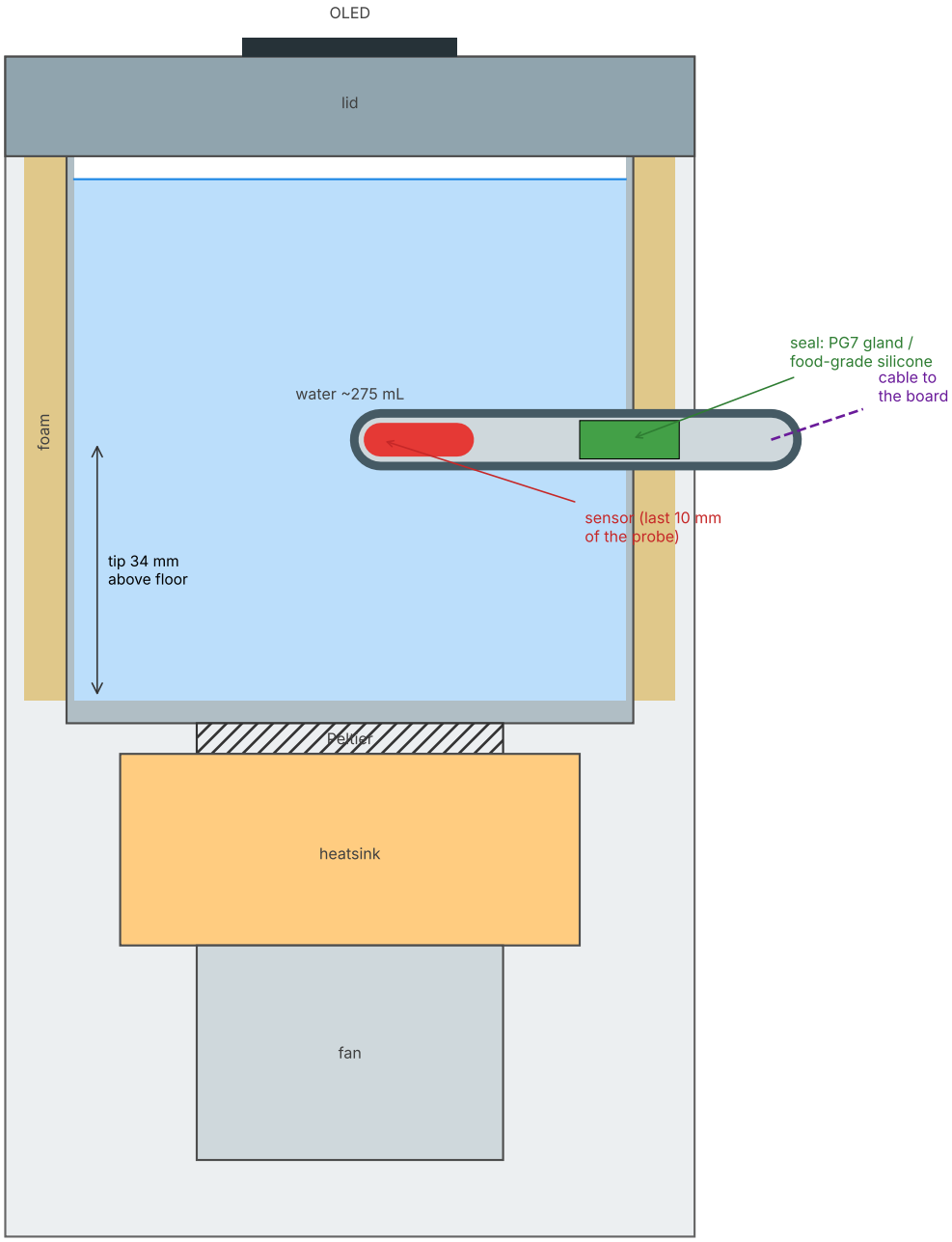
- The angled hole and gland are harder to make
- The lid needs a plug for the sensor cable

Score

Accuracy	*****
No leak	*****
Cleaning / drinking	***..
Cable / lid	***..
Easy to build	***..
Total	**19/25**

Mock design for comparison: sizes from the round bottle (cup 72 mm inside, water 68 mm deep). Not built or tested yet.

DESIGN 4. Side wall, mid-height



What it is

A horizontal tube through the shell, foam and cup wall at half the water depth, tip near the centre.

Good

+ Cable stays in the body, the lid is free
+ Good height in the water

Not so good

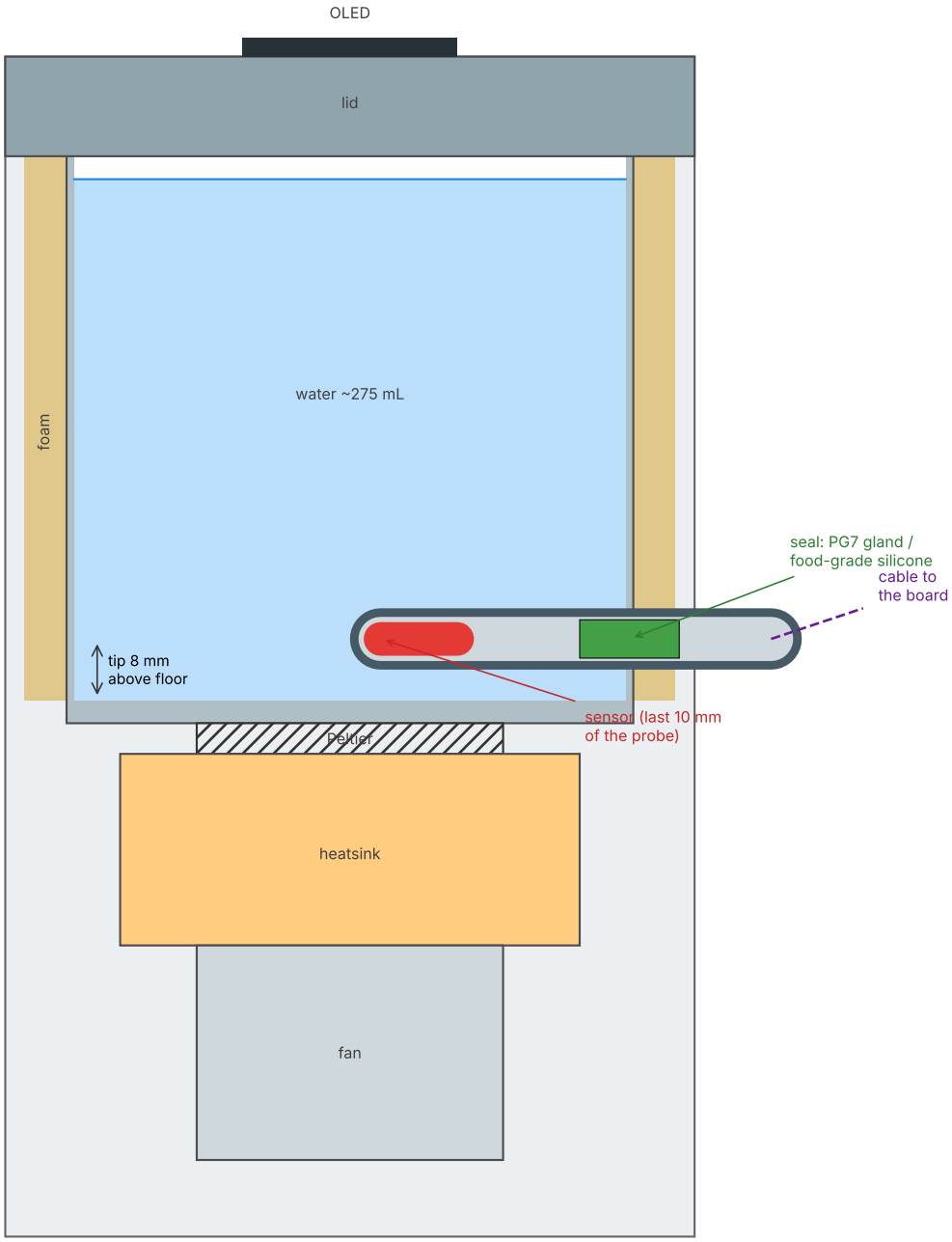
- Hole below the water line: must be sealed and food-safe
- Hard to seal through the 1 mm aluminium wall

Score

Accuracy	****.
No leak	**...
Cleaning / drinking	****.
Cable / lid	*****
Easy to build	**...
Total	**17/25**

Mock design for comparison: sizes from the round bottle (cup 72 mm inside, water 68 mm deep). Not built or tested yet.

DESIGN 5. Side wall, low



What it is

Like design 4, but 8 mm above the floor, close to the Peltier.

Good

+ Cable stays in the body

Not so good

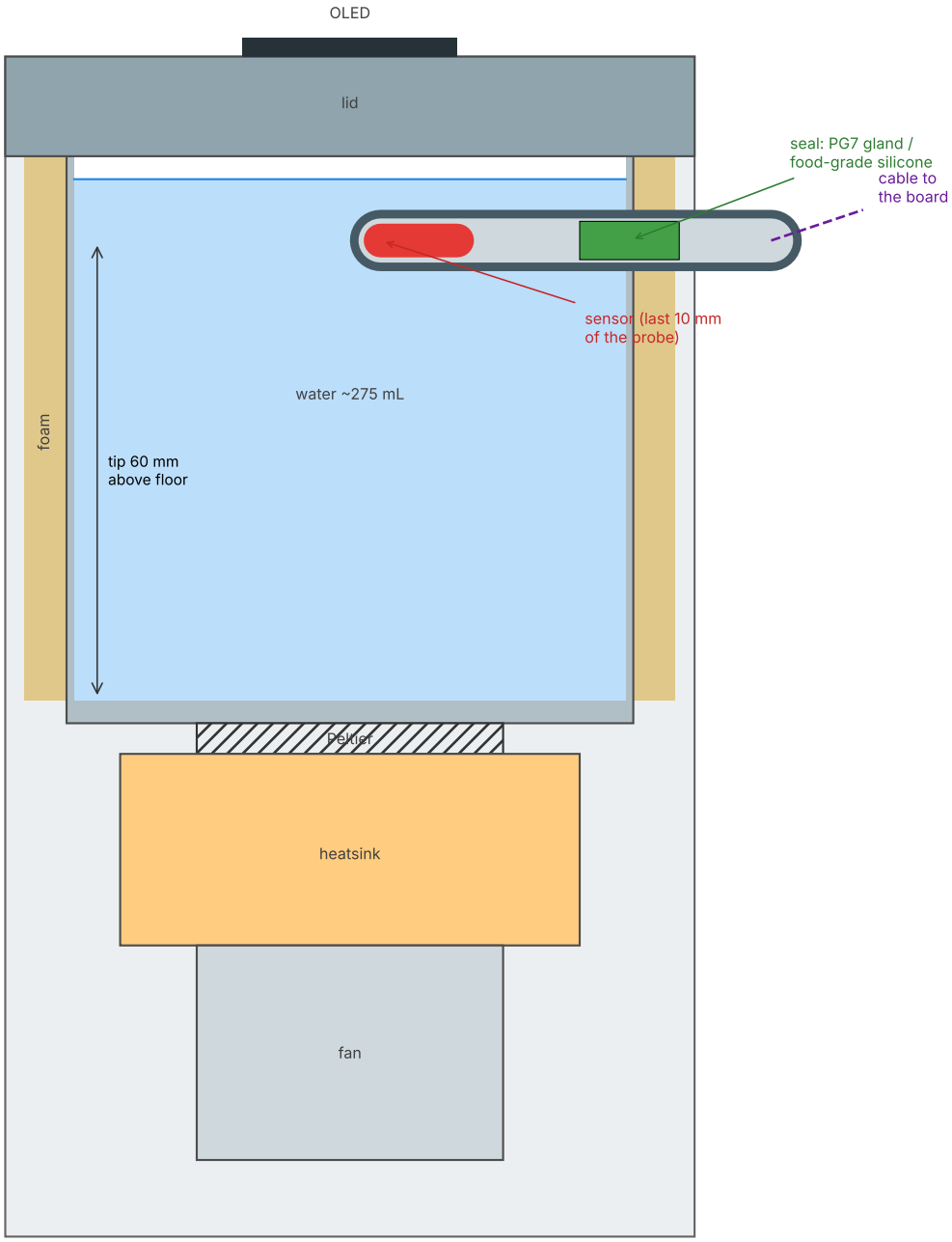
- Reads the water next to the Peltier: the run stops too early
- Hole below the water line, high leak risk

Score

Accuracy	**...
No leak	**...
Cleaning / drinking	****.
Cable / lid	*****
Easy to build	**...
Total	**15/25**

Mock design for comparison: sizes from the round bottle (cup 72 mm inside, water 68 mm deep). Not built or tested yet.

DESIGN 6. Side wall, high



What it is

Like design 4, but 60 mm up, just under the water surface.

Good

+ Cable stays in the body
+ Less water pressure on the seal

Not so good

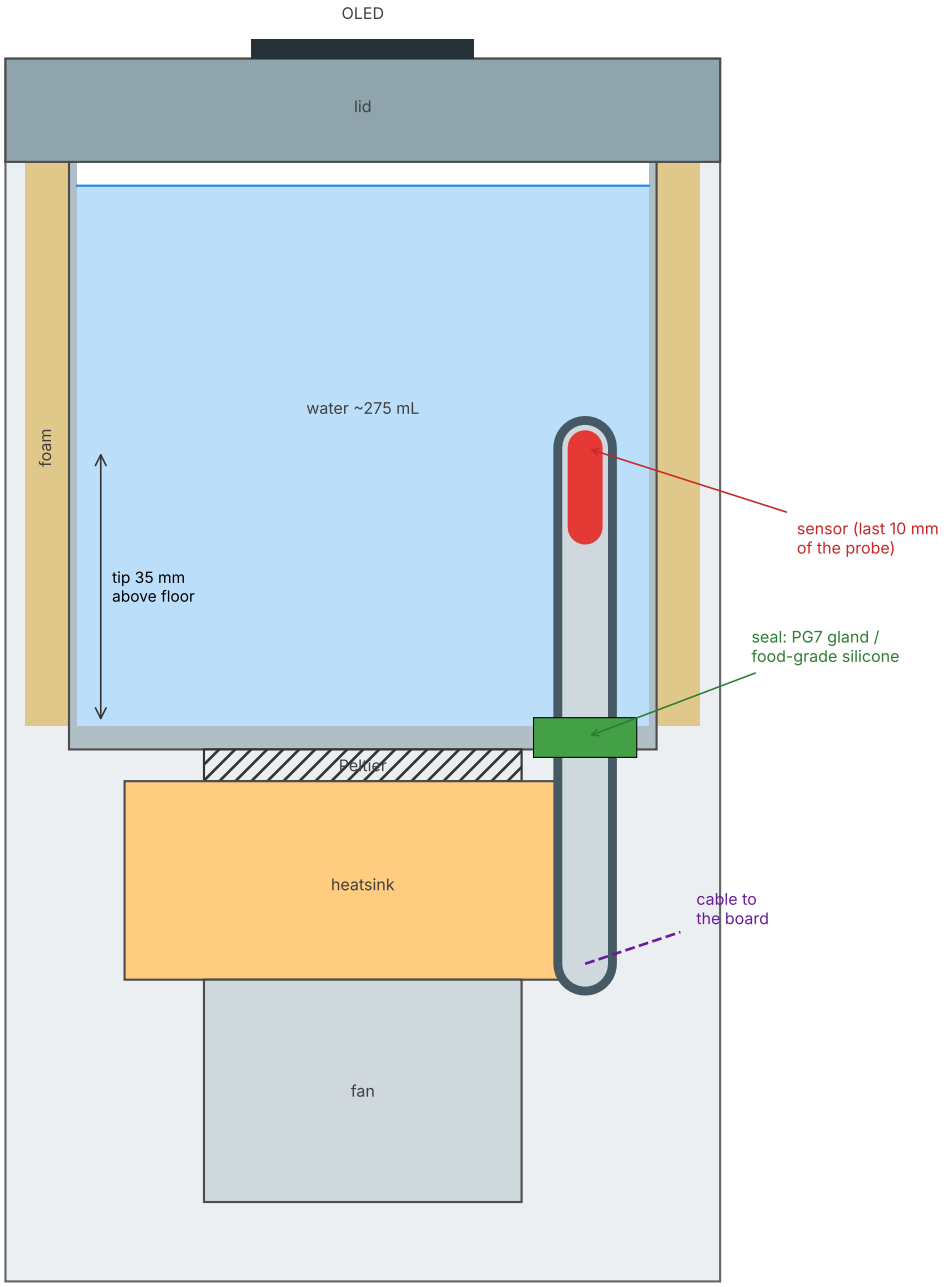
- Reads air when the cup is not full
- The top of the water is not the average

Score

Accuracy	*....
No leak	***..
Cleaning / drinking	****.
Cable / lid	*****
Easy to build	**...
Total	**15/25**

Mock design for comparison: sizes from the round bottle (cup 72 mm inside, water 68 mm deep). Not built or tested yet.

DESIGN 7. Floor, beside the Peltier, straight up



What it is

Comes up through the cup floor beside the 40 mm Peltier, tip at half the water depth.

Good

+ Lid is free, nothing hangs from it

Not so good

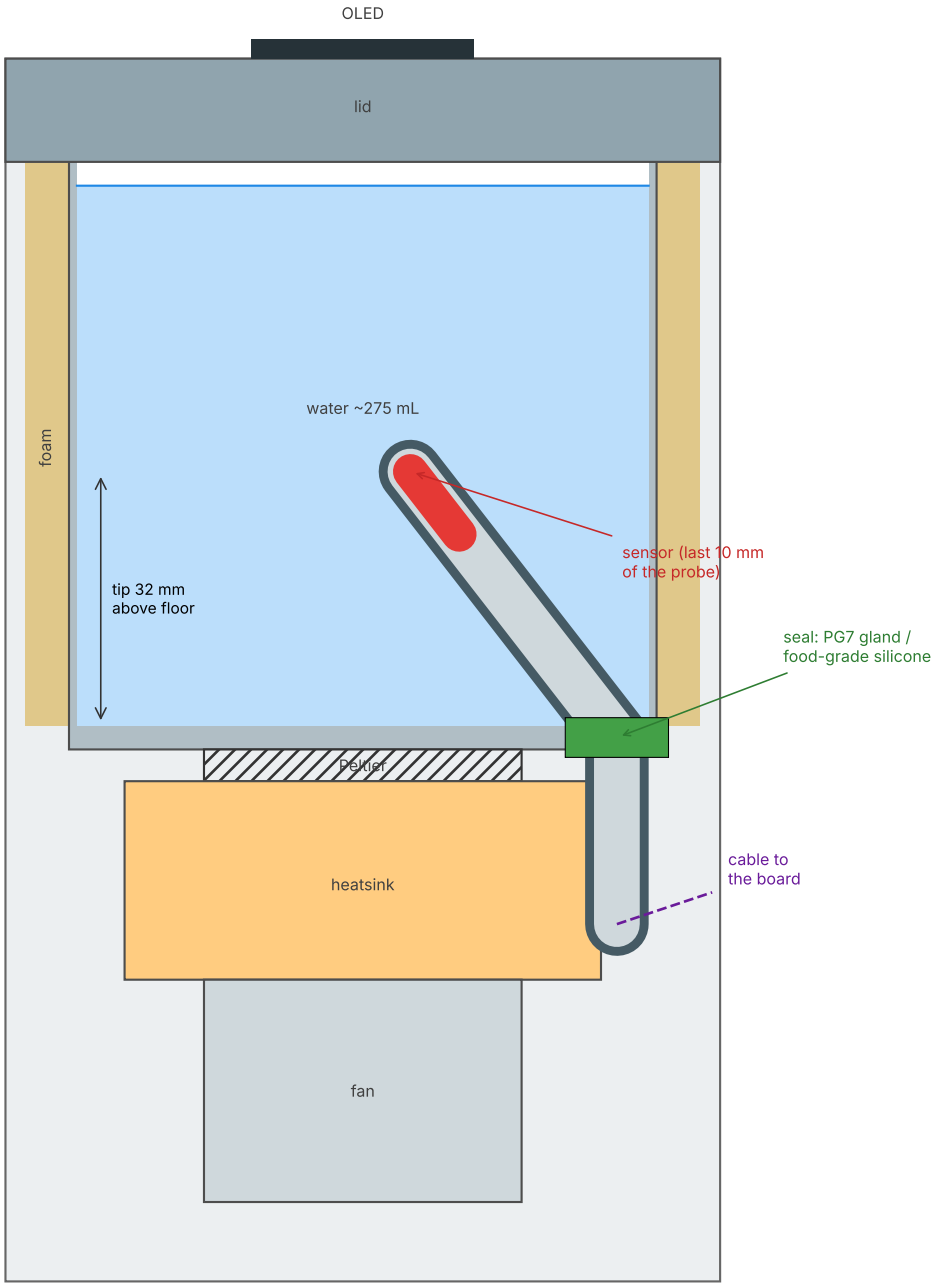
- Hole in the floor right above the electronics: worst leak risk
- Must pass the heatsink and fan area

Score

Accuracy	****.
No leak	*....
Cleaning / drinking	****.
Cable / lid	**...
Easy to build	*....
Total	**12/25**

Mock design for comparison: sizes from the round bottle (cup 72 mm inside, water 68 mm deep). Not built or tested yet.

DESIGN 8. Floor edge, angled in



What it is

Enters at the floor edge and slopes to the centre of the water.

Good

+ Tip in the centre
+ Lid is free

Not so good

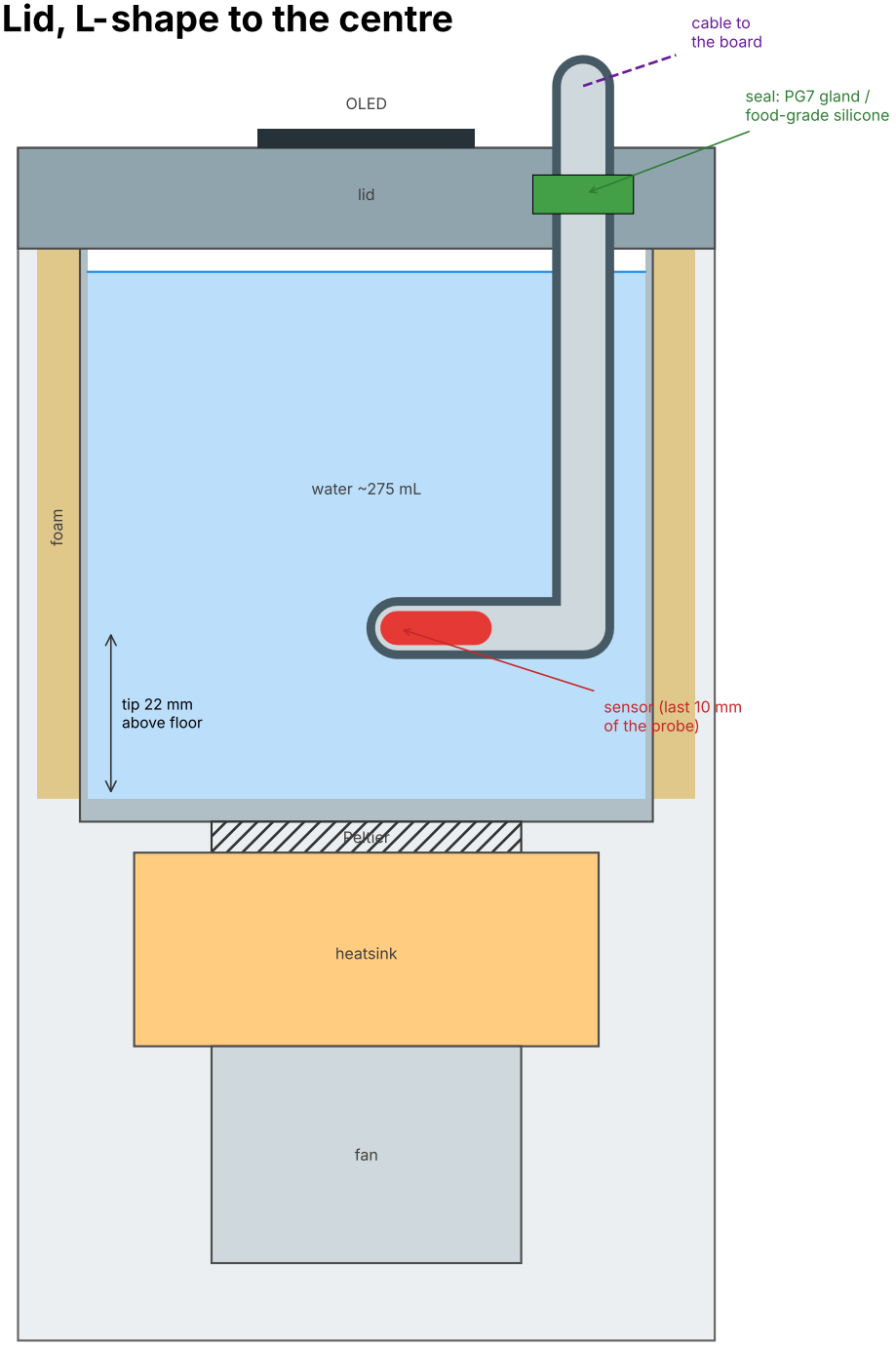
- Hole in the floor above the electronics
- Angled seal in a 3 mm plate is hard to make

Score

Accuracy	****.
No leak	*....
Cleaning / drinking	***..
Cable / lid	**...
Easy to build	*....
Total	**11/25**

Mock design for comparison: sizes from the round bottle (cup 72 mm inside, water 68 mm deep). Not built or tested yet.

DESIGN 9. Lid, L-shape to the centre



What it is

Goes down near the wall, then turns to the centre. The sensor sits in the horizontal part.

Good

+ Tip in the centre at 1/3 height
+ No hole below the water line

Not so good

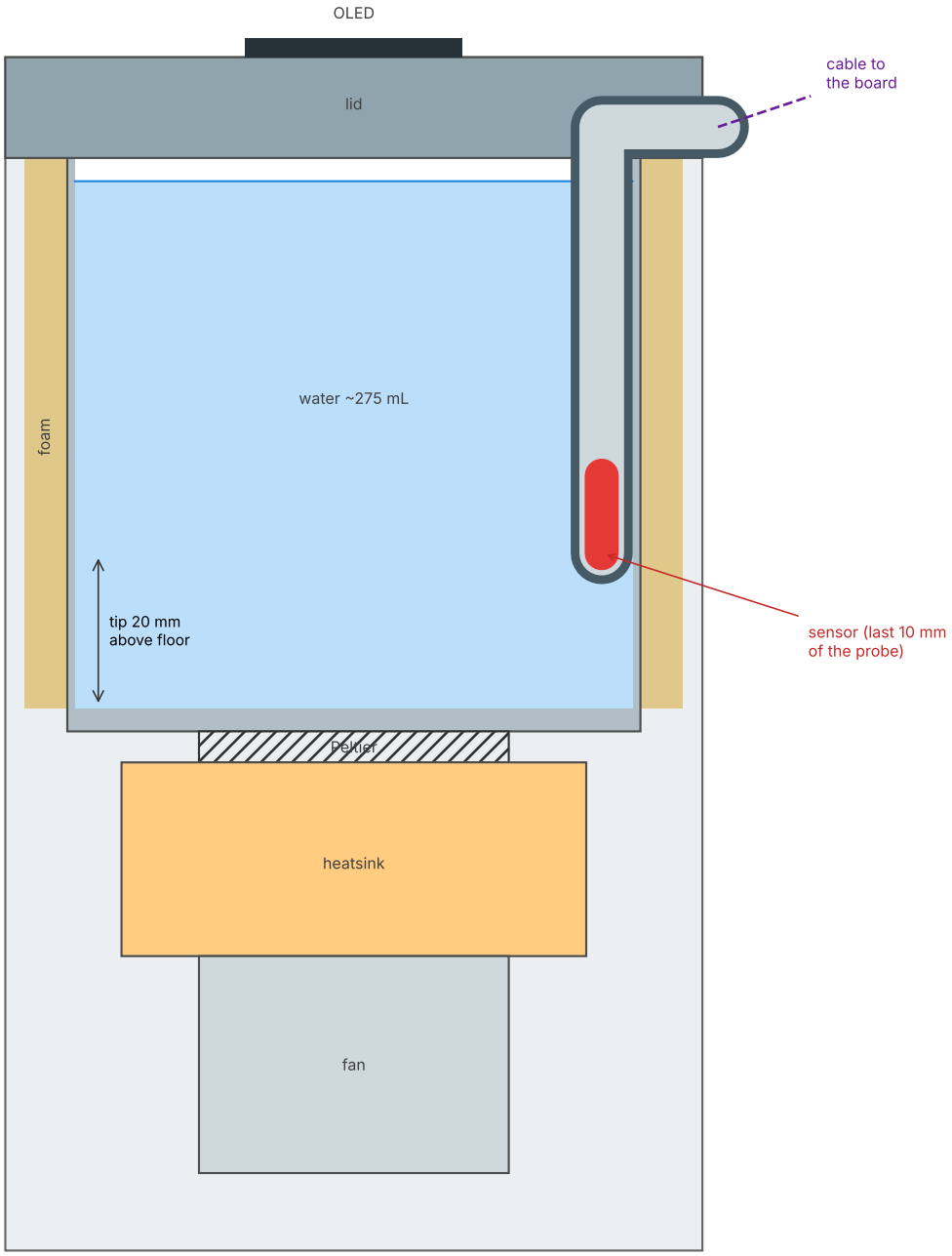
- The probe does not bend: the sensor must be at the tip of a straight part
- Bent tube is in the way of cleaning

Score

Accuracy	*****
No leak	*****
Cleaning / drinking	**...
Cable / lid	***..
Easy to build	**...
Total	**17/25**

Mock design for comparison: sizes from the round bottle (cup 72 mm inside, water 68 mm deep). Not built or tested yet.

DESIGN 10. Over the cup rim, along the inside wall (RECOMMENDED)



What it is

The tube is fixed to the CUP (not the lid): it hooks over the rim and runs down the inside wall to 20 mm above the floor. The cable leaves through a groove in the foam under the lid seal.

Good

+ No hole at all: nothing to seal
+ Lid comes off freely, the cable stays in the body
+ Easy to make: one bent tube and two clips

Not so good

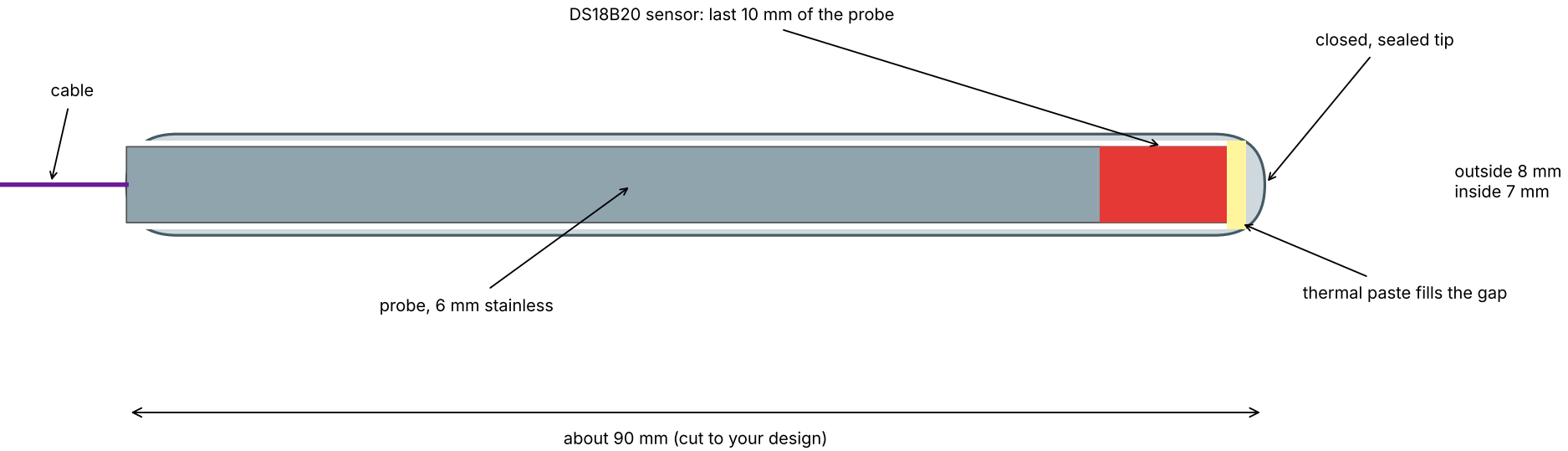
- Tip is 4 mm from the wall, not in the centre
- The lid seal needs a small groove for the tube

Score

Accuracy	****.
No leak	*****
Cleaning / drinking	****.
Cable / lid	*****
Easy to build	****.
Total	**22/25**

Mock design for comparison: sizes from the round bottle (cup 72 mm inside, water 68 mm deep). Not built or tested yet.

HOW TO MAKE THE THERMOWELL



Parts

- Stainless steel tube 304 or 316, 8 mm outside, 7 mm inside (0.5 mm wall), closed at one end.
  Easiest: buy a ready-made 'thermowell 8 mm' (sold for home brewing and smokers). A stainless drinking straw of the right size also works if the end is sealed by a welder (TIG) or with lead-free silver solder.
- Thermal paste (a small tube), food-grade silicone sealant, PG7 cable gland or silicone grommet (lid designs).

Steps

1. Close and check the tip: fill the tube with water and look for drops for 10 minutes. No drops = sealed.
2. Put a little thermal paste in the tube, then push the probe in until it touches the tip.
3. Fix the tube (gland in the lid, or clips over the cup rim for design 10).
4. The tip must sit in the water: about 20 mm above the floor, away from the Peltier and the surface.
5. Hold the cable so it cannot pull the probe out (cable tie or a drop of silicone at the top of the tube).
6. Test: compare with a reference thermometer at 10, 20, 30, 40 and 50 C; measure the response time.

Only stainless steel and food-grade silicone touch the drink. No hot glue, no tape, nothing inside the water but the tube.